Feature-Based Knowledge Distillation for Crack Segmentation

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April 14, 2026

Problem Statement

- Detect cracks in concrete structures
- Applications: bridges, tunnels, buildings
- Challenges:
  - Very thin structures
  - Noise and low contrast
  - Class imbalance

Figure: cracks in concrete structures

Image Segmentation

- Pixel-wise classification
- Input: image
- Output: binary mask (binary segmentation)
- Each pixel labeled as crack / no crack

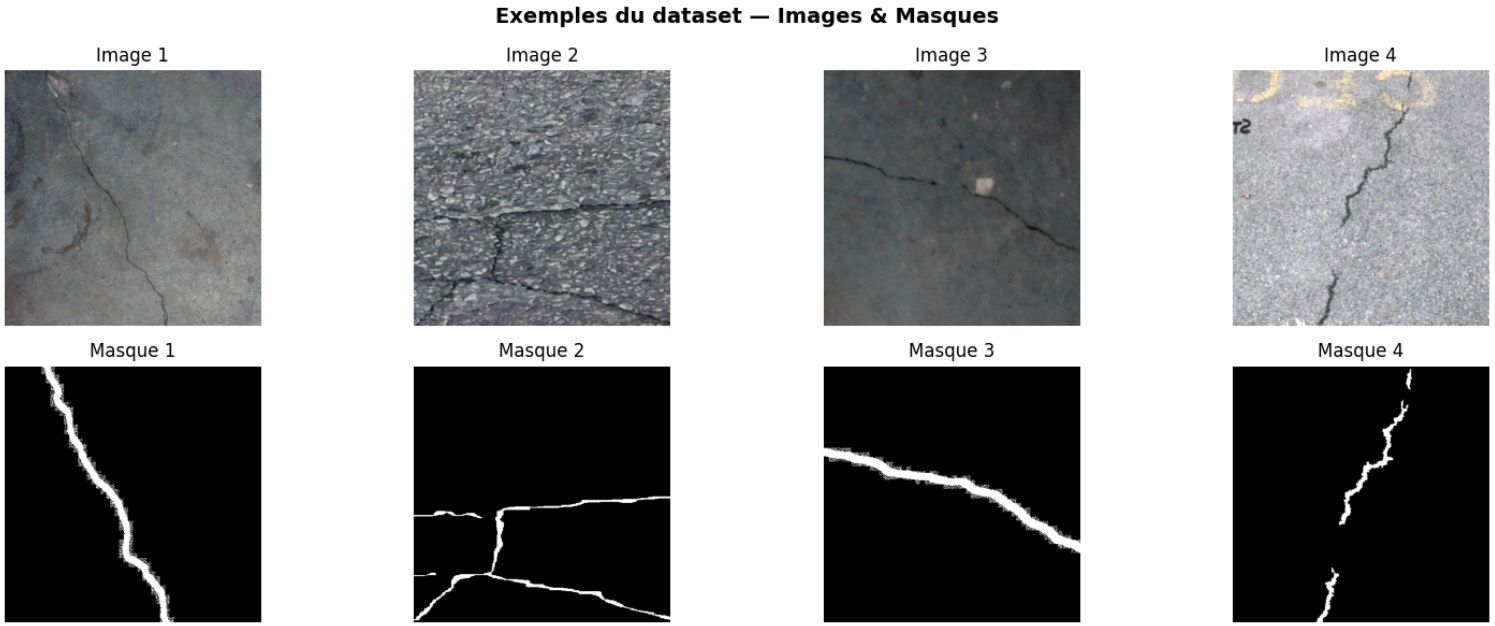


Figure: Example of image and corresponding segmentation mask

U-Net

- CNN architecture for segmentation (6M params)
- Encoder: extracts features
- Bottleneck: middleware
- Decoder: reconstructs segmentation
- Skip connections preserve spatial details

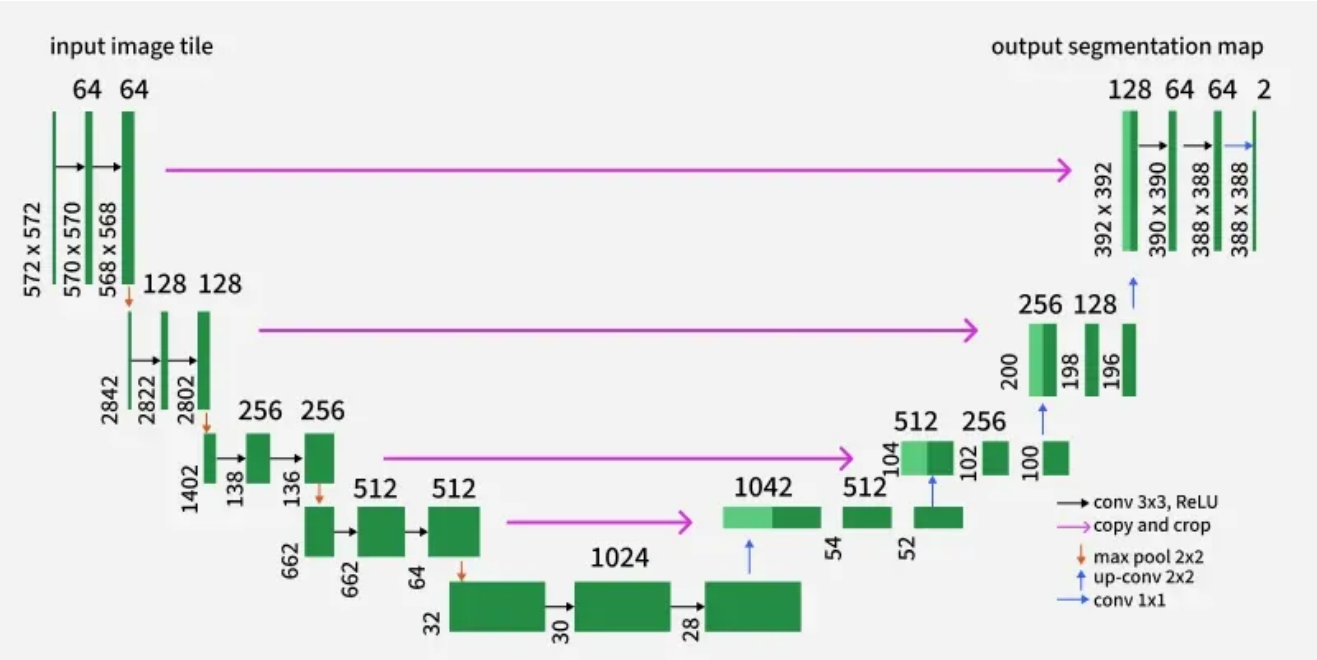


Figure: U-Net architecture

U-Net++

- Improved version of U-Net : Nested skip connections
- Better feature fusion
- Higher accuracy but more complexity (26M params)

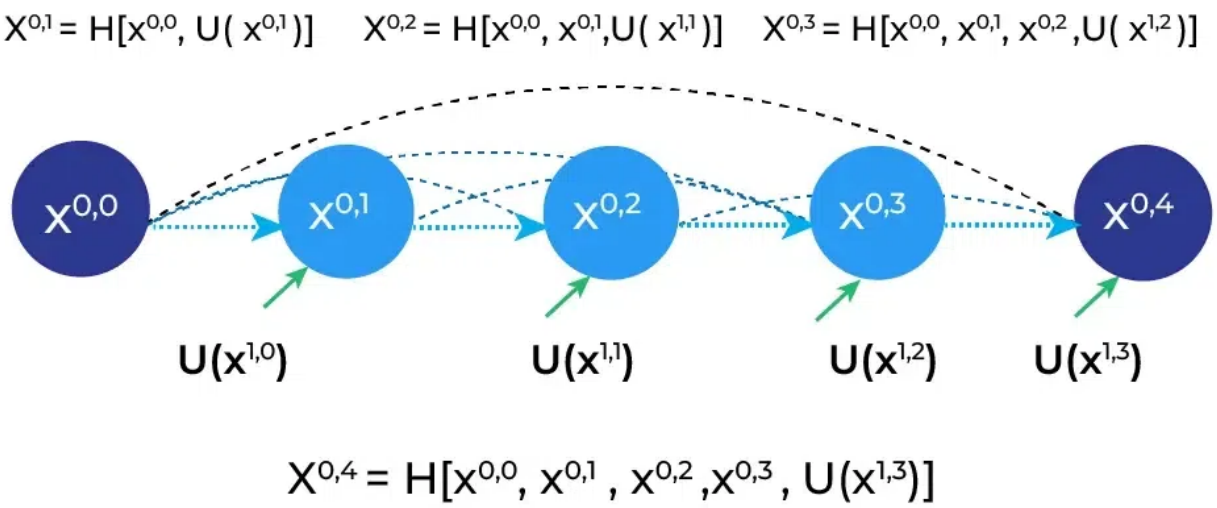


Figure: U-Net++ dense skip pathways

Teacher vs Student

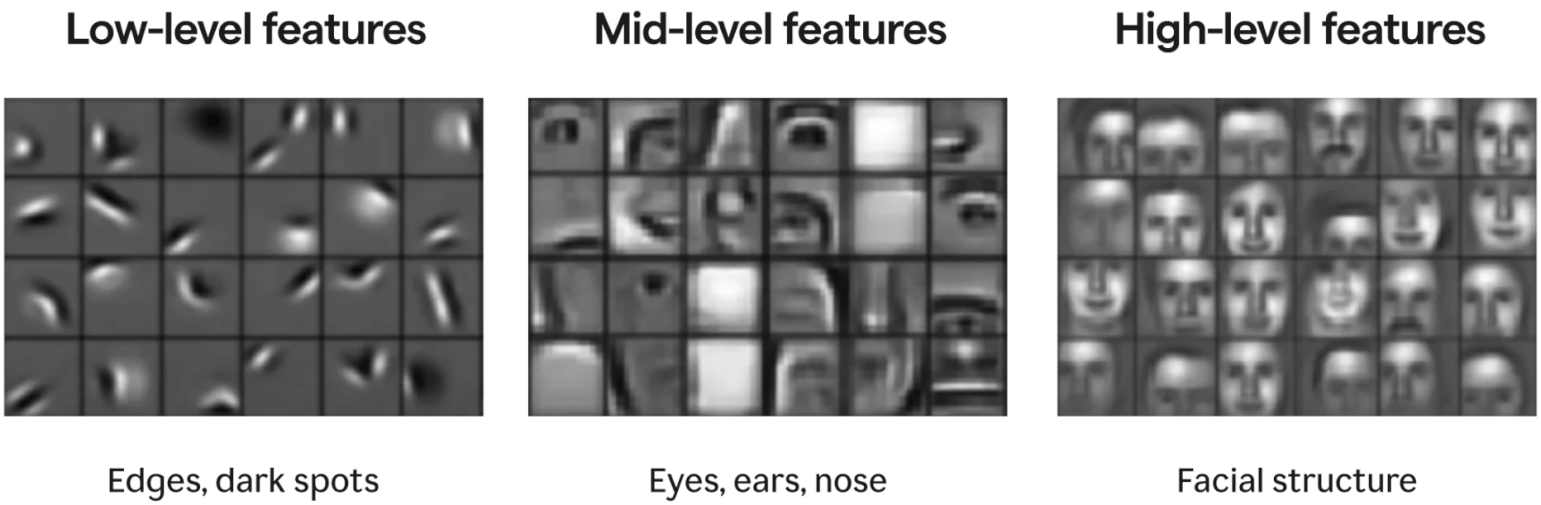
- Teacher:
  - Large model (U-Net++)
  - High accuracy
  - Slow
- Student:
  - Smaller model (Light U-Net)
  - Faster
  - Less accurate

Knowledge Distillation

- Train a small model using a large model
- Student learns from:
  - Ground truth labels
  - Teacher knowledge
- Goal: max performance with min cost

Feature-Based Distillation

- Match internal feature maps
- Not only final predictions
- Helps capture fine details
- Important for micro-crack detection



Training Idea

Combine two losses:

- Segmentation loss
- Distillation loss

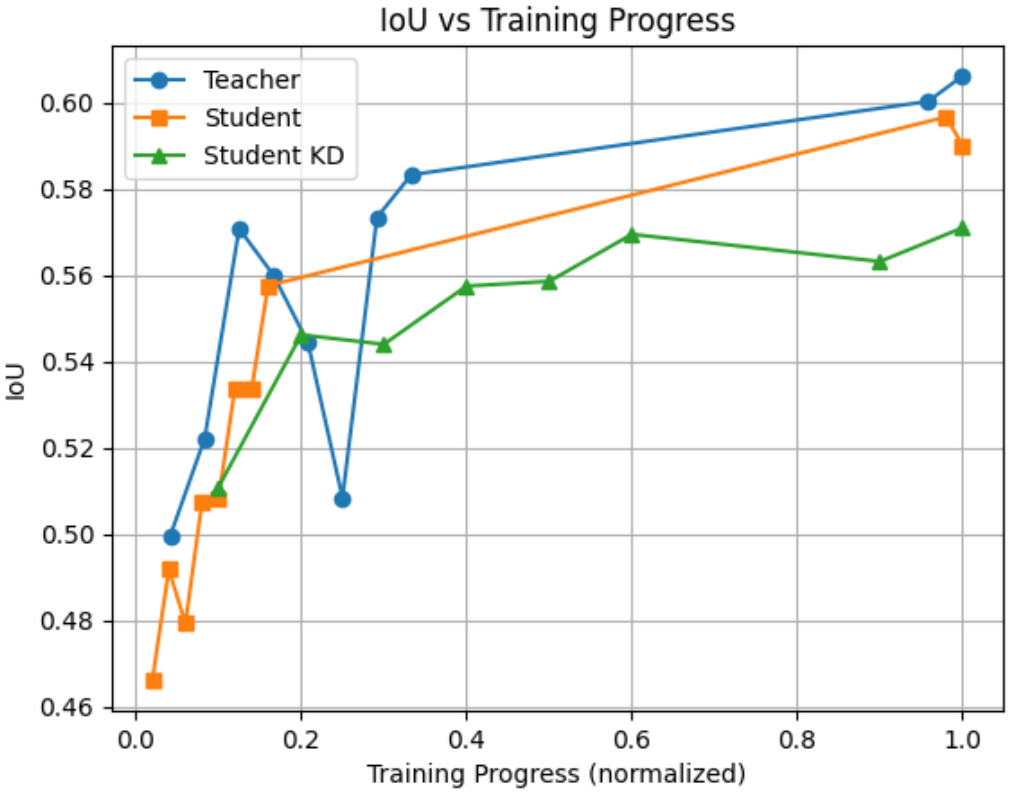
Student learns from:

- Labels
- Teacher features

Evaluation Metrics

- IoU (Intersection over Union)
- F1 Score (Dice)
- Suitable for imbalanced data

IoU Comparison (Segmentation Performance)



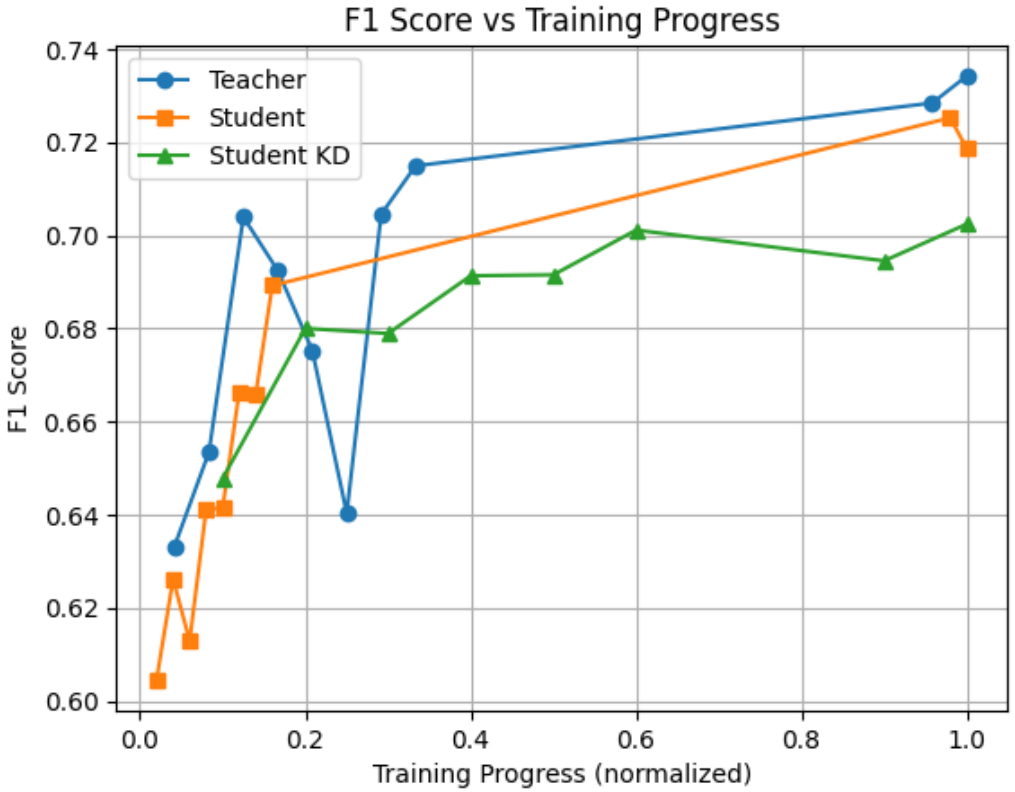
Observations:

- Student alone approaches teacher performance. (no convergence)
- KD reaches similar IoU **earlier**.

Insight:

- KD enables **faster representation learning**.

F1 Score Comparison



Observations:

- Similar trend to IoU.
- KD improves early-stage F1 significantly.
- Final KD performance is slightly lower.

Insight:

- KD impacts **learning speed more than final accuracy**.

Experimental Limitation

Limitation:

- KD model trained for only **10 epochs**.
- Teacher: 24 epochs Student: 50 epochs

Reason:

- Hardware constraints (limited computational resources).

Impact:

- U-Net ++ and KD model did not fully converge.
- Full distillation potential could not be observed.

Expected with More Resources:

- KD would likely approach teacher performance.
- While maintaining faster convergence.

Conclusion

Key Takeaways:

- KD improves **training efficiency**.
- Faster learning in early stages.

Future Work:

- use pre-trained teacher models.
- run distillation longer enough.